

Endogenous Radiolytic Production of Oxygen and Its Potential to Support Habitable Environments in Ocean Worlds

By: Thu Phuong Duong^{1,2}
Supervisor: Dr. Ngoc Truong³

¹University of Science and Technology of Hanoi, Vietnam

²National Taiwan Normal University, Taiwan (*coming soon*)

³Southwest Research Institute, the USA

The 2025 SAGI Astrophysics Workshop

Introduction

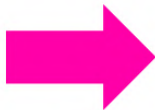


Figure: Earth. (NASA)



Figure: Me. (*for anyone attending online that cannot see me now*)

Life beyond Earth?



Where else in the solar system can we find liquid water?

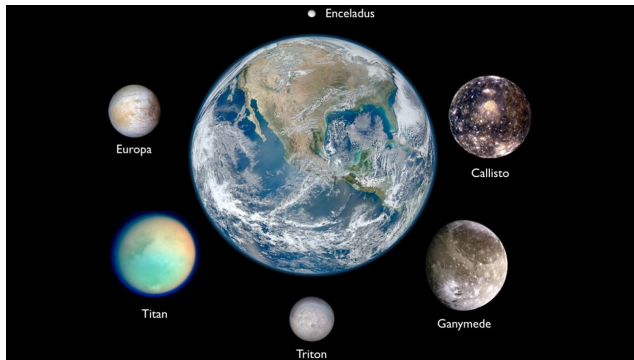


Figure: Some Ocean Worlds. (Schmidt Ocean Institute)



Europa moon



Figure: Portrait of Europa taken by Galileo spacecraft in late 1990s. (NASA/JPL-Caltech/SETI Institute)

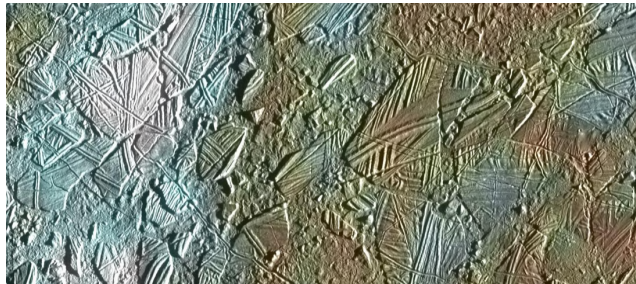
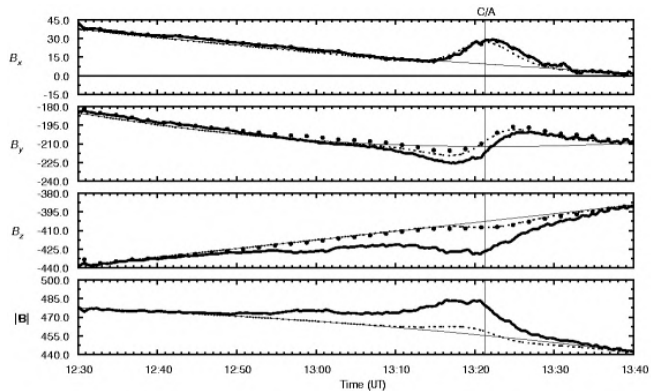


Figure: Zoom in. (NASA/JPL/University of Arizona)

Evidence of a global ocean of salty water



x	-9.84	-8.17	-6.56	-4.95	-3.31	-1.65	0.04	1.75
y	8.06	6.16	4.34	2.51	0.67	-1.16	-2.96	-4.72
z	0.28	0.31	0.34	0.37	0.39	0.41	0.42	0.43
R	12.72	10.24	7.87	5.56	3.40	2.06	2.99	5.05

Figure: Jupiter's disrupted magnetic field around Europa's space, measured by Galileo spacecraft. The observed magnetic field components and magnitude (in nT) are shown as thick solid lines. (*Khurana et al.. 1998*)

Europa moon



Figure: Portrait of Europa taken by Galileo spacecraft in late 1990s. (NASA/JPL-Caltech/SETI Institute)

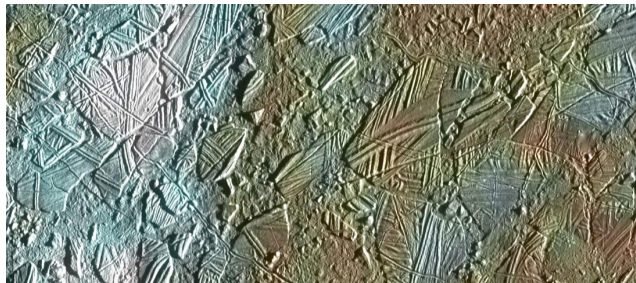


Figure: Zoom in. (NASA/JPL/University of Arizona)

Idea

- Asteroids and comets collided with Europa moon and may have left organic materials. (JPL, 2013)

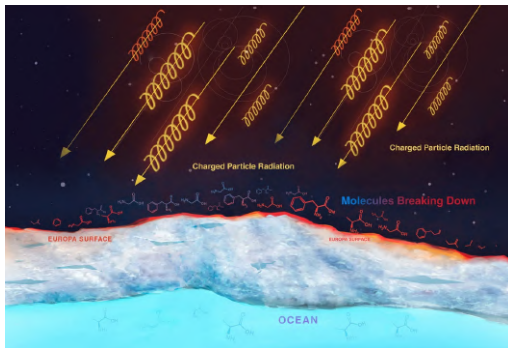
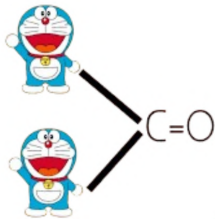


Figure: Europa's surface is blasted by radiation from Jupiter. That's a bad thing for life on the surface – it couldn't survive. But the radiation may create fuel for life in an ocean below the surface. (NASA/JPL-Caltech)

Life's ingredients: Oxygen

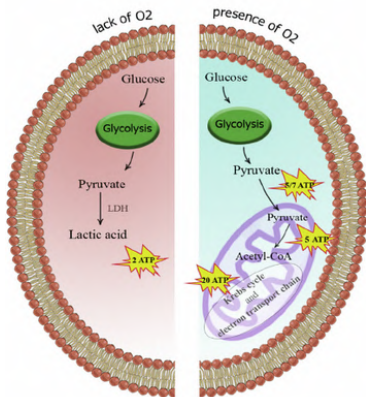


Figure: Difference in glucose metabolic pathways in the presence and absence of oxygen. (Movahed et al., 2019)

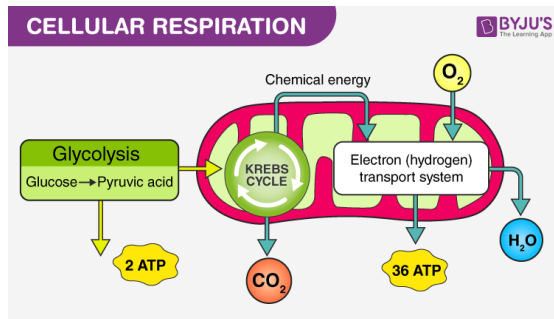
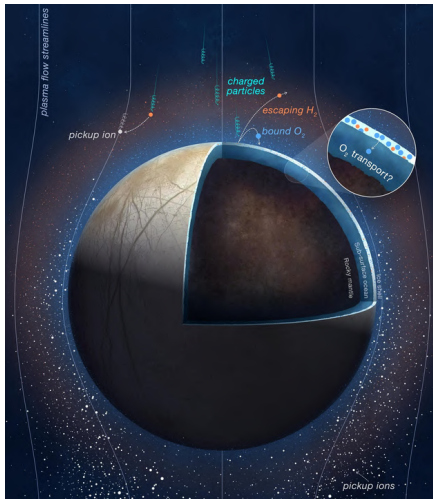


Figure: O_2 helps break down nutrients to release energy.

Could Europa's ocean support life?



- Current idea: Surface radiolysis produce oxygen.
 - Caveat: How do the oxygen get through the $\sim 30 - 35\text{km}$ thick ice **shell** to the ocean?
- In-situ production.

Figure: O_2 production on Europa.
(NASA/JPL-Caltech/SwRI/PU)

Water radiolysis

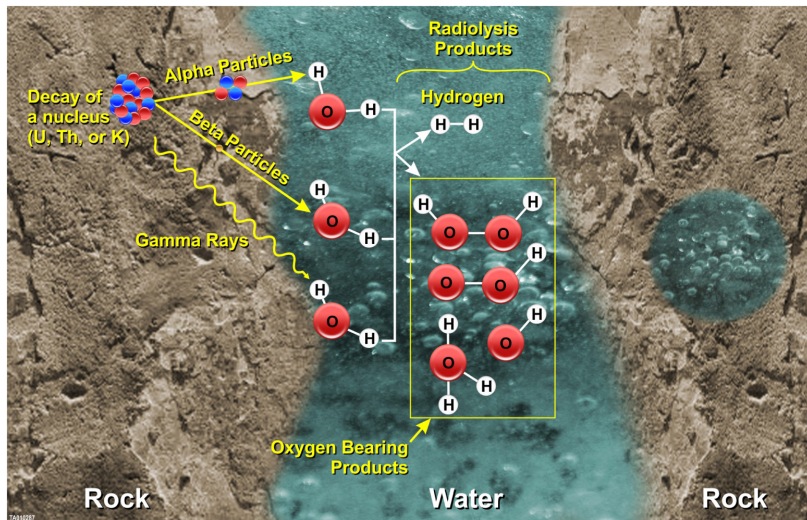


Figure: In-situ O_2 production. (Bouquet et al., 2017)

Water radiolysis

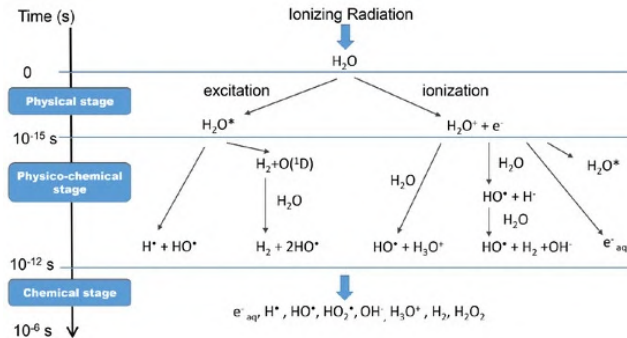


Figure: Events in the radiolysis of water. (Macdonald et al., 2021)

Table: Some reactions lead to O_2 formation.

No.	Reaction
1	$HO_2 + HO_2 \rightarrow H_2O_2 + O_2$
2	$HO_2 + H_2O_2 \rightarrow O_2 + OH + H_2O$
3	$OH + HO_2 \rightarrow O_2 + H_2O$
4	$OH + O_2^- \rightarrow O_2 + OH^-$
5	$HO_2 + O_2^- \rightarrow O_2 + HO_2^-$
6	$O_2^- + O_2^- \rightarrow O_2 + HO_2^- + OH^-$
7	$O_2^- + H_2O_2 \rightarrow O_2 + OH^- + OH$
8	$H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$

Molecular oxygen is not formed as a primary product of water radiolysis, but appears via secondary reactions.

Main questions

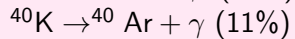
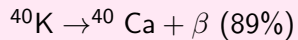
- How much oxygen can be produced through γ emission of ^{40}K decay in Europa's subsurface ocean?
- What are the implications for Europa's potential habitability?



The 2025 SAGI Astrophysics Workshop 14 / 25

Model setup

^{40}K decay



(Ershov, 2020)



Redistribution of 40K on Europa

ϵ_0 : Total extraction factor

ϵ_1 : Partial extraction factor

ϵ_2 : **Ocean-only factor**

ϵ_{1a} : Refined model derived from ϵ_1 and adjusted based on observed Na/K ratios in Europa's atmosphere (Brown, 2001)

Assumption: Ratio of radioactive potassium to total potassium is constant and similar to Earth's $\sim 0.0119\%$. (Amiro, 2003)

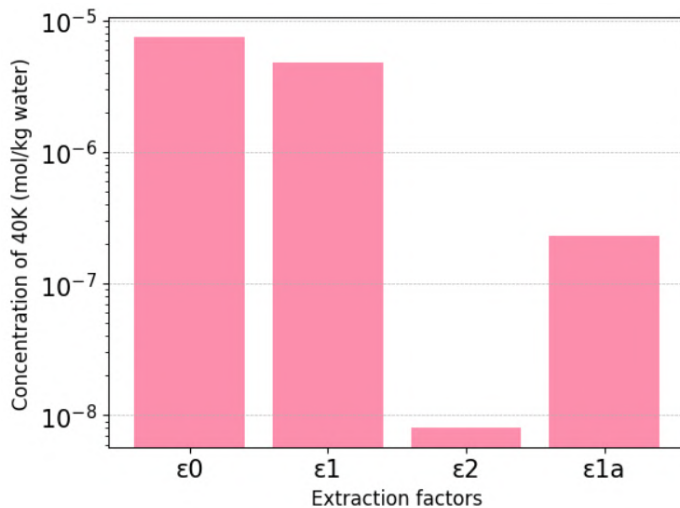


Figure: Molality concentration of 40K.

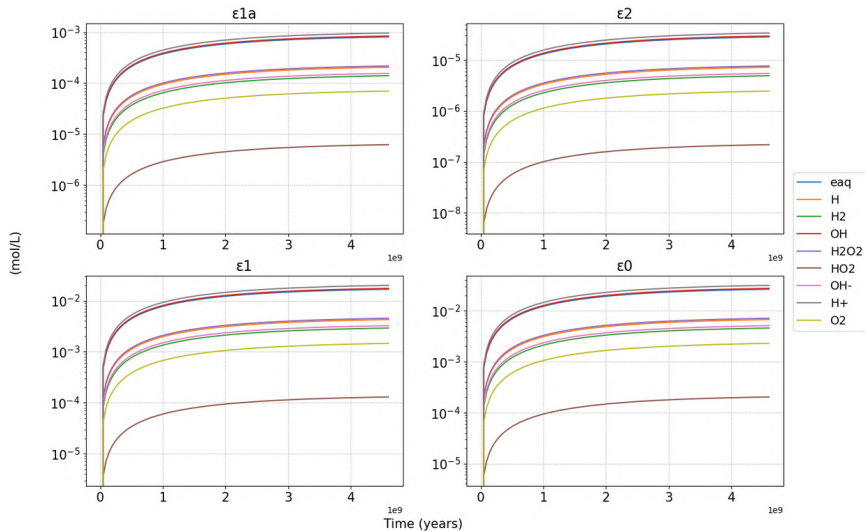
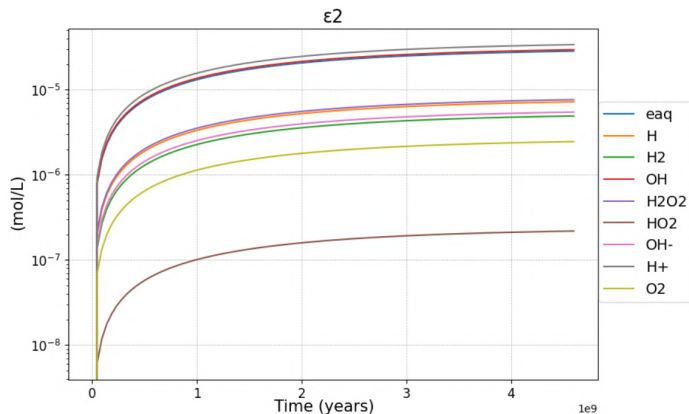


Figure: Concentration of some radiolysis products.



- Most abundant: H^+ , OH , H_2O_2 , OH^- , ...
- O_2 are less abundant.

Figure: Concentration of some radiolysis products.

Organisms in the ocean

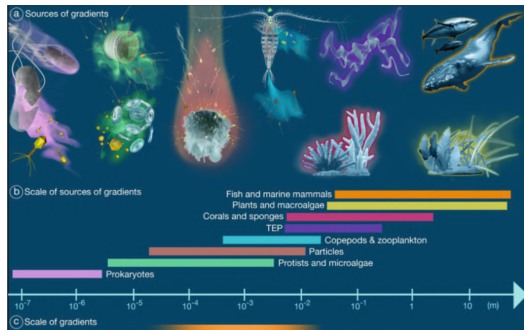


Figure: Some different organisms in the ocean and their scales. (*Clerc et al., 2022*)

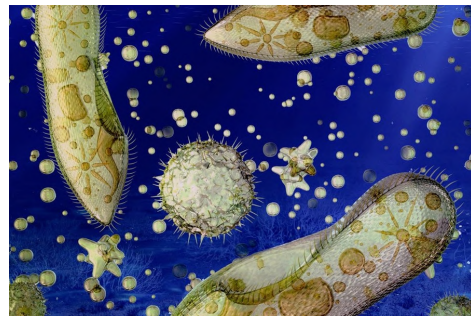


Figure: MIT oceanographers have found that some extremophiles, mostly bacteria in OMZs, have a surprisingly low limit of O_2 needed. (*MIT News*)

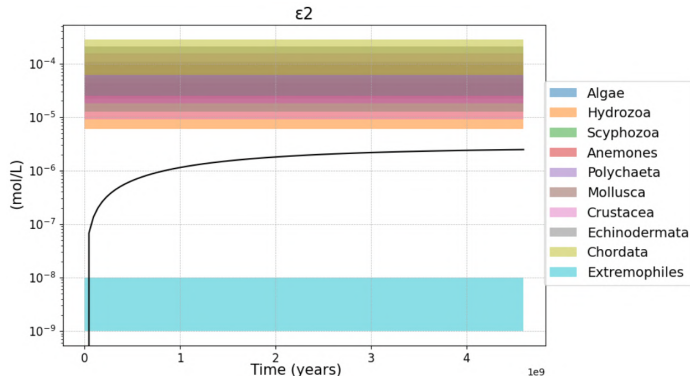
Thresholds for microbial survival

Table: Median sublethal oxygen concentrations of benthic marine organisms and extremophiles.

Taxa	Sublethal oxygen threshold (mg/L)	
	Min.	Max.
Algae	2.96	2.96
Hydrozoa	0.19	1.73
Scyphozoa	0.4	0.57
Anemones	0.71	1.43
Polychaeta	0.57	3.43
Mollusca	0.57	3.24
Crustacea	0.29	5
Echinodermata	0.79	6.63
Chordata	1.94	9
Extremophiles	0.000032	0.00032



Discussion



- Results exceed the minimum oxygen requirements for extremophiles on Earth.
- Radiolysis can be recognized as an **important energy source** for potential life on Europa.



How much oxygen can be produced through γ emission of ^{40}K decay in Europa's subsurface ocean?

- The γ emission of ^{40}K decay can generate **micromolar-level concentrations** of O_2 in Europa's ocean, which exceeds the minimum oxygen requirements for aerobic extremophiles on Earth ($\sim 1 \text{ nM}$).

What are the implications for Europa's potential habitability?

- Radiolysis **may** represent a significant and previously underappreciated energy source for potential life in Europa's ocean, though further data is needed to confirm this hypothesis.

Key takeaways:

- 40K decay generates **micro-level concentrations** of oxygen which exceeds the minimum oxygen thresholds for extremophiles on Earth.
- Extremophiles' ability to adapt suggests endogenous radiolysis **could** be a crucial source for possible life on Europa.
- This topic is an **ongoing debate** and warrants further study.
- Future mission Europa Clipper (next 5–6 years) with direct Europa measurements helps refine models and improve our understanding of life beyond Earth.



Thank you for not
sleeping when I talk
listening!

For further discussion:
duongphuong2109@gmail.com



Many thanks to Dr. Ngoc for guiding me !!!